

VELSK, Russian Federation

WMO: 228670

Lat: 61.0693N

Lon: 42.0695E

Elev: 91

StdP: 100.24

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.5	(c) -27.9	(d) -34.4	(e) 0.1	(f) -31.4	(g) -30.8	(h) 0.2	(i) -27.9	(j) 5.9	(k) -4.8	(l) 5.2	(m) -4.3	(n) 0.2	(o) 50	(p) 0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 28.9	(d) 19.7	(e) 26.8	(f) 18.8	(g) 24.9	(h) 17.5	(i) 21.0	(j) 26.7	(k) 19.7	(l) 25.1	(m) 18.7	(n) 23.4	(o) 2.0	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.8	23.7	17.8	12.9	22.3	16.8	12.1	21.4	61.0	26.6	56.9	25.2	53.4	23.4	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.4	4.0	-33.9	31.4	4.5	2.0	-37.2	32.9	-39.8	34.0	-42.3	35.1	-45.6	36.5		
			-34.1	22.3	4.5	1.5	-37.3	23.4	-39.9	24.2	-42.4	25.1	-45.7	26.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-11.6	-10.3	-4.3	3.1	10.0	14.8	17.9	14.9	9.7	3.2	-3.9	-8.2		
	DBStd	11.60	8.59	8.07	5.36	4.83	5.21	4.47	3.87	3.87	3.82	4.25	6.42	7.90		
	HDD10.0	3221	670	568	445	211	67	8	1	4	51	215	416	564		
	HDD18.3	5669	928	802	703	456	263	124	56	118	261	471	666	822		
	CDD10.0	671	0	0	0	6	67	153	245	157	41	3	0	0		
	CDD18.3	77	0	0	0	0	3	18	42	13	0	0	0	0		
	CDH23.3	773	0	0	0	2	51	199	392	128	2	0	0	0		
	CDH26.7	194	0	0	0	0	8	52	103	32	0	0	0	0		
(14) Wind	WSAvg	1.6	1.6	1.7	1.8	1.7	1.7	1.4	1.2	1.2	1.5	1.9	1.8	1.9		
(15) Precipitation	PrecAvg	584	36	29	32	34	52	66	70	75	49	53	48	41		
	PrecMax	687	59	61	79	108	90	164	163	144	114	102	88	60		
	PrecMin	453	12	10	5	7	24	12	18	21	21	4	11	22		
	PrecStd	68	11	13	17	21	18	34	34	33	23	24	18	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	3.0	9.2	21.6	27.8	31.0	32.2	30.3	23.2	16.1	8.1	5.4		
		MCWB	1.2	2.0	3.9	12.7	16.3	20.0	21.4	20.8	17.3	11.7	7.1	4.3		
	2%	DB	1.3	1.7	6.1	17.2	24.5	28.0	29.4	26.5	19.7	12.6	5.7	2.9		
		MCWB	0.6	0.8	2.3	9.7	14.9	18.4	20.1	19.3	15.0	10.2	4.8	2.1		
	5%	DB	0.6	0.9	3.8	13.5	21.9	25.5	27.4	24.1	17.5	10.6	4.4	1.6		
		MCWB	0.0	0.1	1.1	7.4	13.3	17.1	19.5	17.7	13.7	9.0	3.7	0.8		
	10%	DB	-0.7	-0.2	2.5	10.9	19.4	23.2	25.4	21.8	15.6	9.0	3.0	0.8		
		MCWB	-1.3	-1.0	0.5	6.0	12.2	16.2	18.6	16.7	12.8	7.7	2.4	0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	2.3	4.6	12.9	18.2	21.1	22.8	22.3	18.1	12.6	7.3	4.3		
		MCDB	1.7	2.8	7.9	20.6	24.3	28.3	29.4	28.4	21.8	15.0	8.1	5.3		
	2%	WB	0.7	0.9	2.8	10.2	16.3	19.7	21.4	19.9	15.6	10.6	4.9	2.1		
		MCDB	1.1	1.5	4.9	15.5	22.1	26.0	27.2	24.8	18.6	12.3	5.5	2.8		
	5%	WB	0.0	0.1	1.7	8.3	14.4	18.4	20.4	18.5	14.3	9.2	3.7	0.9		
		MCDB	0.7	0.8	3.4	12.7	20.0	23.7	25.7	22.8	16.8	10.4	4.3	1.5		
	10%	WB	-1.4	-1.0	0.8	6.4	12.9	17.0	19.4	17.4	13.1	7.7	2.4	0.1		
		MCDB	-0.8	-0.2	2.1	10.4	18.4	22.0	24.3	20.9	15.2	8.9	3.0	0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.1	8.6	9.7	10.8	10.3	10.3	9.2	7.5	4.5	4.1	5.0		
		MCDBR	4.8	4.8	9.3	14.7	14.7	13.9	13.2	12.8	10.9	7.4	4.1	3.6		
	5% WB	MCWBR	4.8	4.4	6.3	8.4	6.8	5.8	5.6	5.9	6.6	5.3	3.9	3.6		
		MCDWR	4.8	4.6	7.0	13.3	12.7	12.1	11.7	10.6	9.4	7.0	4.0	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.302	0.329	0.369	0.356	0.354	0.390	0.375	0.342	0.321	0.279	0.226		
		taud	2.027	2.067	2.090	2.203	2.433	2.463	2.374	2.395	2.474	2.384	2.134	1.971		
	Edn at Noon		449	648	771	817	868	875	827	810	778	652	421	310		
		Edh at Noon	42	77	106	116	100	99	106	96	75	58	38	26		
(44) All-Sky Solar Radiation	RadAvg		0.28	0.90	2.15	3.62	4.95	5.19	5.23	3.73	2.12	0.85	0.25	0.13		
	RadStd		0.05	0.24	0.36	0.41	0.34	0.49	0.42	0.35	0.25	0.12	0.04	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.73	N/A	N/A	N/A	N/A	N/A	-222	-253	N/A	N/A
(47) Regional (5 neighbors)		+0.67	N/A	N/A	N/A	N/A	N/A	-195	-234	N/A	N/A

Nomenclature: See separate page